

Trends

“Increasing Demand For ADVANCED INSPECTION TOOLS In Electronics, Automotive, And Medical Industries”

What is fuelling the surge in demand for advanced optical inspection and measurement systems? Which new features are being introduced to support evolving use cases? To explore these questions, Akanksha Sondhi Gaur from EFY spoke with Steve Sanderson of Vision Engineering. Here is what emerged...



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What is a modular lens setup?

A modular lens setup allows users to switch between different optics based on specific requirements. Advanced solutions now support auto-sensing lens technology, meaning that when a new lens is attached, the system automatically detects it and adjusts the settings accordingly. Vision Engineering's lenses are engineered for precision imaging, offering minimal distortion, high contrast, and excellent depth of field.

How do such systems handle difficult lighting conditions?

High-quality camera systems include both automatic and manual exposure controls. They incorporate spot exposure functionality, enabling the camera to adjust exposure for a designated region of interest. This feature is particularly advantageous for subjects with highly reflective surfaces or complex lighting environments. Users can also choose motion priority to track moving subjects or depth priority to optimise aperture settings for superior depth of field management.

What is built-in illumination, and how does it help?

Unlike earlier models with fixed built-in illumination, newer systems offer interchangeable lighting modules. This modular approach allows users to swap different illumination sources depending on the application. Available options include quadrant illumination, compact spot lighting, and combined white and UV lighting for fluorescence applications. The ability to adjust and customise lighting ensures optimal image capture across varied use cases.

How does angled lighting impact glare, reflections, and overall imaging performance?

Strategically placing lighting at different angles can significantly reduce glare and unwanted reflections. Large-area soft lighting helps diffuse direct reflections, improving image clarity and reducing visual artefacts. Proper lighting control enhances contrast and prevents excessive light scatter, which is especially critical in microscopy for capturing fine details with high accuracy.

What are the advantages of an eyepiece-less microscope?

Eyepiece-less microscopes, such as Vision Engineering's Lynx Evo, provide an ergonomic viewing experience by eliminating the need for traditional eyepieces. This design improves operator comfort, enabling a neutral neck position and reducing eye strain. Users can work with high precision over extended periods—a crucial advantage for detailed tasks. The system is also highly modular, offering various configurations, ring lights, and stands to suit different applications.

How do such microscopes assist in quality control during the assembly process?

These microscopes can be seamlessly integrated into the assembly line, allowing operators to inspect components at multiple stages. Since many final products conceal their internal components, in-line

This interview is an excerpt of a longer discussion. For the full transcript, please visit: <https://efy.co.in/interviewjune25>

